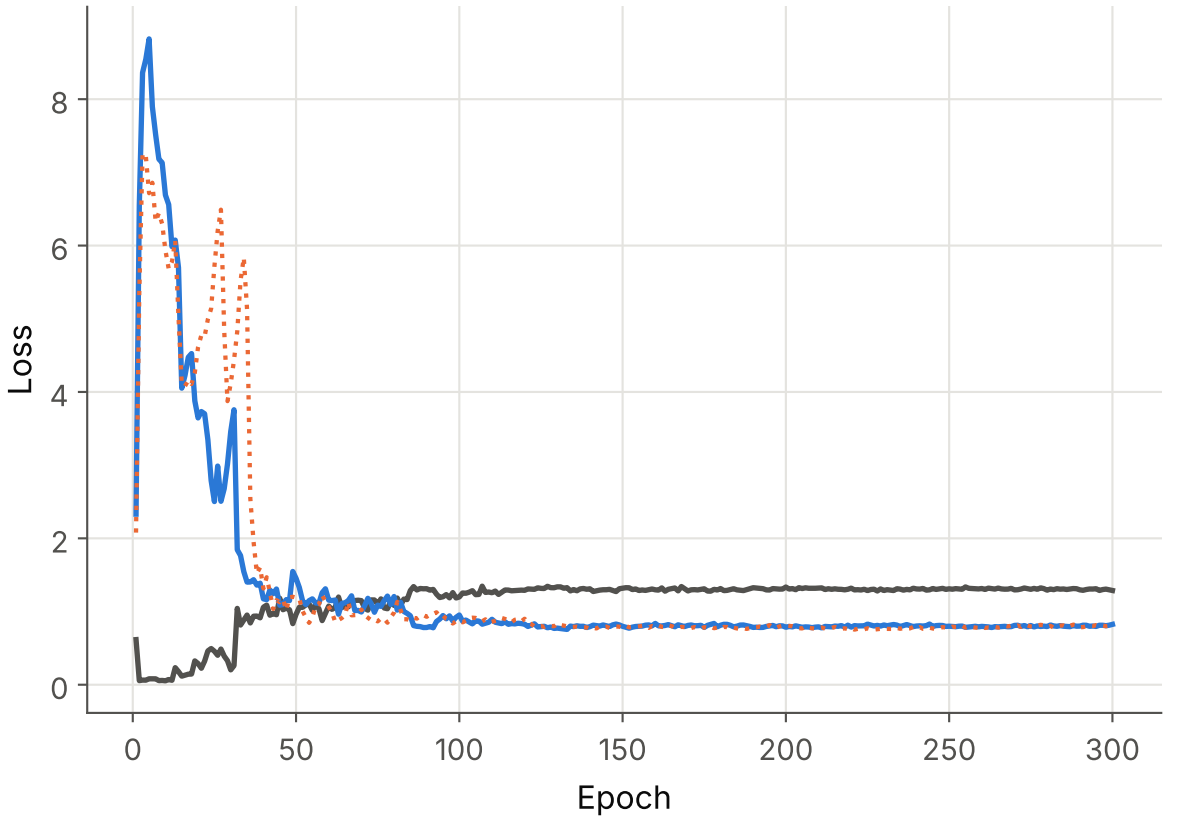
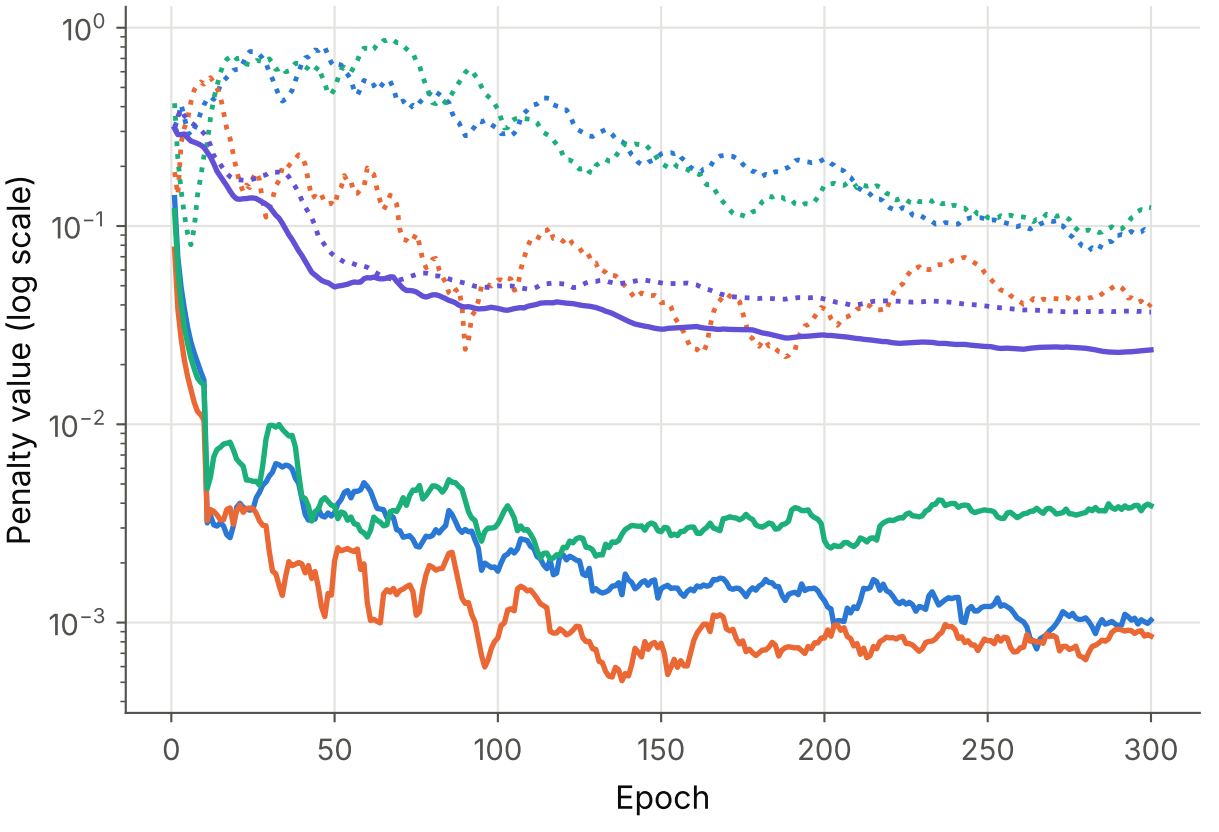
(a) DWCA: adversarial dynamics (seed 0)



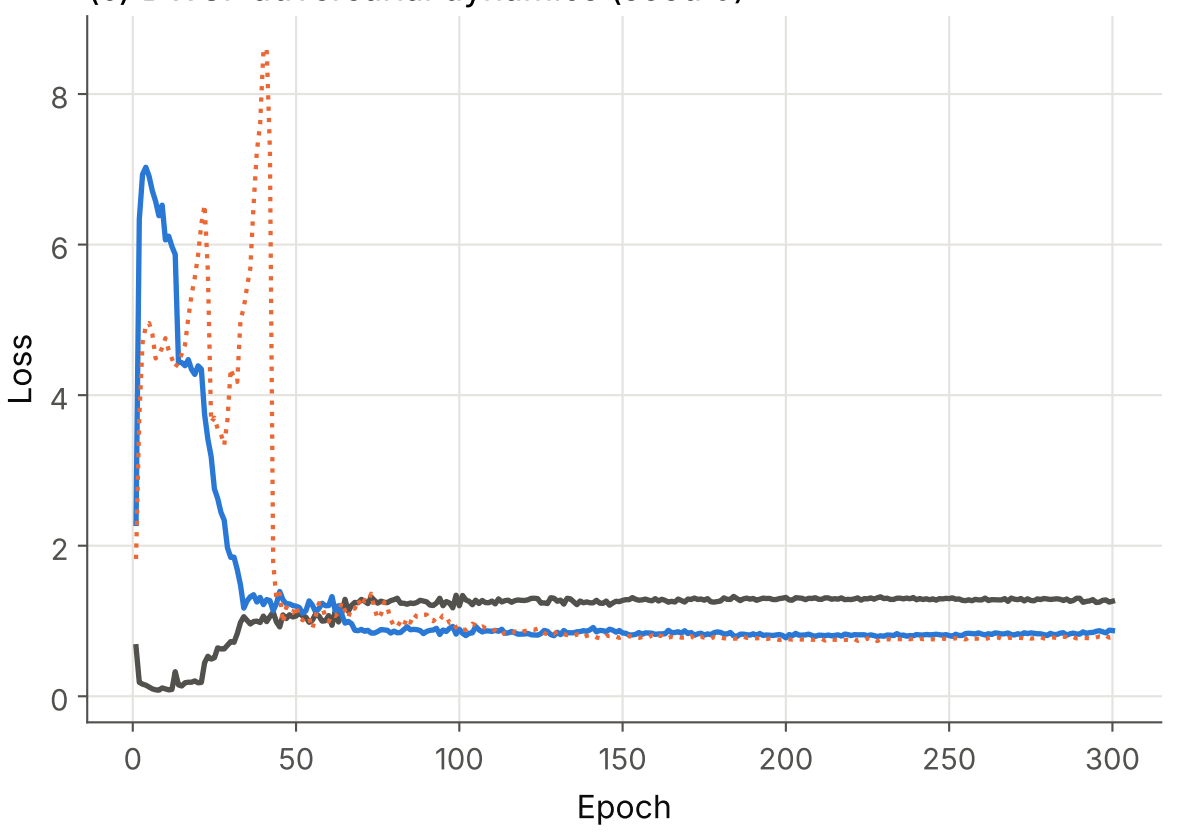
(b) DWCA: petrophysical penalty terms (10-epoch mean)



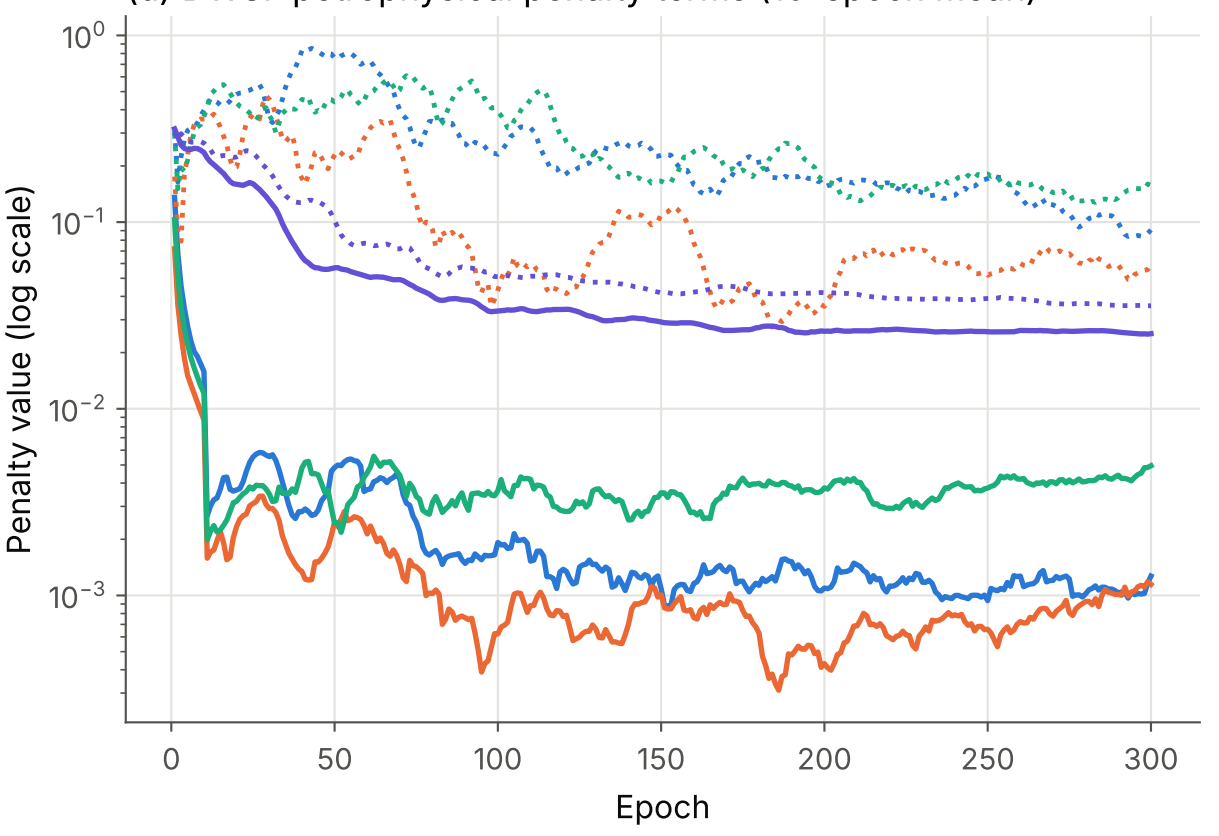
Discriminator loss
Generator adversarial loss (ChemoGAN-H)
Generator adversarial loss (unconstrained GAN)

Corr. NPHI–RHOB
Corr. NPHI–DT
Corr. PE–DWCA
TV smoothness
ChemoGAN-H (solid)
Unconstrained GAN, not optimised (dotted)

(c) DWSI: adversarial dynamics (seed 0)



(d) DWSI: petrophysical penalty terms (10-epoch mean)



Discriminator loss
Generator adversarial loss (ChemoGAN-H)
Generator adversarial loss (unconstrained GAN)

Corr. NPHI–RHOB
Corr. NPHI–DT
Corr. PE–DWSI
TV smoothness
ChemoGAN-H (solid)
Unconstrained GAN, not optimised (dotted)